

National Institute of Business Management School of Computing and Engineering Course work Assessment Announcement Sheet			
Course Name	Higher National Diploma in Software Engineering		
Module Name	Mobile Application Development		
Batch	25.2FT		
Learning Outcomes Covered (Mention according to the Module Descriptor)	1. Apply Language for application development.		
	2. Familiarize with Development Tools.		
	3. Implement known UI component to own application.		
	4. Create a high standard application and publish it.		
Assesement CW No			
Assesement Mode	Individual Group	Group (if it is group mode only)	
		Group Size	Grouping Criteria
		Maximum of 4 Members	Students are allowed to select their group members according to their preferences
Assesement Type	Practical Test Report Software Presentation VIVA MCQ		
	If other specify		
Hand in Date Time			
Hand out Date Time			
Submission Details (Format and Location)	Submit the source code as .zip file to the LMS on or before the deadline. If the file is too large to be uploaded to LMS, upload it to OneDrive, Google Drive or GitHub and share the project link by submitting a text file to LMS clearly mentioning group details.		
Plagiarism Criteria			
Assesement CW Description			
Assignment: Android Application Development using Kotlin Overview The objective of this assignment is to design and develop a fully functional, production-ready Android mobile application using Kotlin. This project will test your ability to integrate hardware features, persistent storage, and network communication while maintaining a high standard of User Experience (UX). Technical Requirements Your application must satisfy the following technical components: A. User Interface (UI/UX) Responsive: The app must handle different screen sizes and orientations. Usability: Implementation of intuitive navigation and clear user. B. Local Database Integration Persistence: Use Room Persistence Library to store structured data locally. Operations: Must perform full CRUD (Create, Read, Update, Delete) operations.			

- **Offline Support:** The app should remain functional (viewing existing data) without an active internet connection.

C. Hardware Access

- **Camera:** Implement image capture functionality (e.g., for profile pictures, scanning, or documentation).
- **Location:** Access GPS coordinates to provide context-aware features or mapping.
- **Permissions:** Properly handle runtime permissions for both Camera and Location.

D. Remote API Connectivity / Data source

- **Networking:** Integrate an API or remote data source like firebase.
- **Data Handling:** Fetch, parse (JSON), and display remote data within the UI.
- **Concurrency:** Use Kotlin Coroutines for non-blocking network calls.

Task

Consider the following options and select one of the following scenarios or propose your own scenario with a similar weight in the scope.

Scenario 1: Agri Scout (Field Crop Disease & Yield Tracker)

Agricultural extension officers and farm managers often work in rural or deep-field locations where cellular networks are unreliable or non-existent. They need a tool to document crop health, identify diseases, and estimate yields in the field, ensuring data isn't lost when disconnected. The application should contain the following functionalities:

- The application must allow officers to log farm details, crop types, and observations completely offline. Data must be saved locally and queued for a background sync worker when the network is restored.
- Users must be able to capture high-resolution images of diseased leaves or pests. The app must use the device's GPS to automatically append exact geolocation coordinates to these images so outbreaks can be mapped.
- When online, the app must authenticate with a central agricultural backend (REST API/Firebase DB) to push queued field reports and pull-down updated disease identification catalogs or weather warnings.

Scenario 2: "Retail Route" (Field Sales & Van Sales Manager)

Sales representatives who manage distribution to local retail shops and merchants need to process orders, manage van inventory, and issue receipts rapidly. They cannot afford to halt the checkout process in a shop just because the internet connection drops. The application should implement the following functionalities:

- The application must download the daily product catalog, pricing tiers, and the assigned client route at the start of the day. Reps must be able to create invoices and deduct from van inventory entirely offline.
- The app requires the camera to act as a barcode scanner for quickly adding products to an order. It must also connect via Bluetooth to a portable thermal printer to issue physical receipts to shop owners on the spot.
- Sync end-of-day sales data back to the company's central Data source or API, process digital payments made via external gateways, and reconcile inventory discrepancies.

Scenario 3: Asset Guard (Small Business Inventory)

Most construction companies often struggle with "asset leakage" the loss of tools, equipment or electronics due to poor tracking. Create an android application to track physical assets across multiple sites. The application should contain the following functionalities:

- User Management & authentications
- Facilitate the user by adding new assets along with their images, barcodes/QR codes.

- A searchable directory of all company equipment, including serial numbers, purchase dates, and current location/site.
- A workflow to assign assets to specific employees or projects by scanning barcodes or QR codes of the assets.
- A dedicated mode for performing "asset auditing" where users verify the physical presence of an item. System records the precise time and place of the verification.
- Management reports

Notes:

- Identify the requirements of the selected scenario and submit a project proposal to get approval from your lecturer before starting the development.
- You are expected to add any additional features or enhancements that you think by focusing on the completeness of the application.
- Ensure that your app is well-tested and free of critical bugs before submission.
- Implement appropriate security measures to protect user data and ensure that only authorized users can access specific features.

Submission Requirements

- **Source Code:** Submit the source code of your project as a zip file or Git repository
- **Presentation:** Prepare a presentation to demonstrate the functionality of your application and explain the design choices you made.

Marking Rubric

Marks Range	Front-End Development + UI/UX Design (15%)	Local Database Implementation – Room/SQLite (10%)	Remote DB / API Integration – Firebase/REST (10%)	Hardware Integration (Camera/ Location/ QR) (10%)	Completeness of the Project (5%)	Presentation and Viva (10%)
0% – 39%	No proper UI structure. Poor navigation.	Incomplete or incorrect DB implementation. No proper schema design.	No remote integration or non-functional API/Firebase usage.	Hardware features not implemented or non-functional.	Application incomplete and major features missing.	Poor understanding and poor communication skills
40% – 49%	Basic UI screens with minimal interaction.	Basic DB implementation with limited entities and relationships.	Basic Firebase/API usage with limited data sync.	Camera or location implemented but limited functionality.	Less than half of required features completed.	Average understanding and communication skills

50% – 59%	Properly structured UI and navigation. Acceptable UX.	Proper schema design with relationships and CRUD operations implemented correctly.	Functional Firebase/API integration with proper data syncing.	Camera/Location/QR features implemented correctly with proper permissions handling.	More than half of required features completed successfully.	Good understanding and communication skills
70% and Above	Complete, modern, responsive UI. Clean navigation, state handling, error/loading states properly managed.	Well-designed DB schema with normalization, relationships, DAO best practices, and offline-first handling.	Robust Firebase/API integration with authentication, real-time updates, proper error handling and secure data flow.	Correct and efficient hardware usage (Camera, Location, QR scanning). Accurate time/location recording. Proper runtime permission handling.	All required features implemented and fully functional including edge cases and validations.	Excellent understanding and communication skills